

Research Notes on Medical image computing

Medical image computing

>> Section 1

Diffusion MRI is a structural magnetic resonance imaging modality that allows measurement of the diffusion process of molecules. Diffusion is measured by applying a gradient pulse to a magnetic field along a particular direction. In a typical acquisition, a set of uniformly distributed gradient directions is used to create a set of diffusion weighted volumes. In addition, an unweighted volume is acquired under the same magnetic field without application of a gradient pulse. As each acquisition is associated with multiple volumes, diffusion MRI has created a variety of unique challenges in medical image computing. In medicine, there are two major computational goals in diffusion MRI:

>> Section 2

Image registration is a process that searches for the correct alignment of images. In the simplest case, two images are aligned. Typically, one image is treated as the target image and the other is treated as a source image; the source image is transformed to match the target image. The optimization procedure updates the transformation of the source image based on a similarity value that evaluates the current quality of the alignment. This iterative procedure is repeated until a (local) optimum is found. An example is the registration of CT and PET images to combine structural and metabolic information (see figure). Image registration is used in a variety of medical applications:

>> Section 3

Statistical analysis Statistical methods combine the medical imaging field with modern computer vision, machine learning and pattern recognition. Over the last decade, several large datasets have been made publicly available (see for example ADNI, 1000 functional Connectomes Project), in part due to collaboration between various institutes and research centers. This increase in data size calls for new algorithms that can mine and detect subtle changes in the images to address clinical questions. Such clinical questions are very diverse and include group analysis, imaging biomarkers, disease phenotyping and longitudinal studies.

>> Section 4

Data forms Medical image computing typically operates on uniformly sampled data with regular x-y-z spatial spacing (images in 2D and volumes in 3D, generically referred to as images). At each sample point, data is commonly represented in integral form such as signed and unsigned short (16-bit), although forms from unsigned char (8-bit) to 32-bit float are not uncommon. The particular meaning of

the data at the sample point depends on modality: for example a CT acquisition collects radiodensity values, while an MRI acquisition may collect T1 or T2-weighted images. Longitudinal, time-varying acquisitions may or may not acquire images with regular time steps. Fan-like images due to modalities such as curved-array ultrasound are also common and require different representational and algorithmic techniques to process. Other data forms include sheared images due to gantry tilt during acquisition; and unstructured meshes, such as hexahedral and tetrahedral forms, which are used in advanced biomechanical analysis (e.g., tissue deformation, vascular transport, bone implants).

>> Section 5

Segmentation is the process of partitioning an image into different meaningful segments. In medical imaging, these segments often correspond to different tissue classes, organs, pathologies, or other biologically relevant structures. Medical image segmentation is made difficult by low contrast, noise, and other imaging ambiguities. Although there are many computer vision techniques for image segmentation, some have been adapted specifically for medical image computing. Below is a sampling of techniques within this field; the implementation relies on the expertise that clinicians can provide.

>> Section 6

Visualization plays several key roles in medical image computing. Methods from scientific visualization are used to understand and communicate about medical images, which are inherently spatial-temporal. Data visualization and data analysis are used on unstructured data forms, for example when evaluating statistical measures derived during algorithmic processing. Direct interaction with data, a key feature of the visualization process, is used to perform visual queries about data, annotate images, guide segmentation and registration processes, and control the visual representation of data (by controlling lighting rendering properties and viewing parameters). Visualization is used both for initial exploration and for conveying intermediate and final results of analyses. The figure "Visualization of Medical Imaging" illustrates several types of visualization: 1. the display of cross-sections as gray scale images; 2. reformatted views of gray scale images (the sagittal view in this example has a different orientation than the original direction of the image acquisition; and 3. A 3D volume rendering of the same data. The nodular lesion is clearly visible in the different presentations and has been annotated with a white line.

>> Section 7

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Image-based physiological modelling Traditionally, medical image computing has seen to address the quantification and fusion of structural or functional information available at the point and time of image acquisition. In this regard, it can be seen as quantitative sensing of the underlying anatomical, physical or physiological processes. However, over the last few years, there has been a growing interest in the predictive assessment of disease or therapy course. Image-based modelling, be it of biomechanical or physiological nature, can therefore extend the possibilities of image computing from a descriptive to a predictive angle. According to the STEP research roadmap, the Virtual Physiological Human (VPH) is a methodological and technological framework that, once established, will enable the investigation of the human body as a single complex system. Underlying the VPH concept, the International Union for Physiological Sciences (IUPS) has been sponsoring the IUPS Physiome Project for more than a decade,. This is a worldwide public domain effort to provide a computational framework for understanding human physiology. It aims at developing integrative models at all levels of biological organization, from genes to the whole organisms via gene regulatory networks, protein pathways, integrative cell functions, and tissue and whole organ structure/function relations. Such an approach aims at transforming current practice in medicine and underpins a new era of computational medicine. In this context, medical imaging and image computing play an increasingly important role as they provide systems and methods to image, quantify and fuse both structural and functional information about the human being in vivo. These two broad research areas include the transformation of generic computational models to represent specific subjects, thus paving the way for personalized computational models. Individualization of generic computational models through imaging can be realized in three complementary directions: